

Chapter 1 Smooth Manifold

1.1 Definitions

Definition 1.1.0.1 An n -dimensional **smooth manifold** is a topological manifold $(M, \{(U_i, \varphi_i)\})$ with **smooth transition function**

$$\varphi_i \circ \varphi_j^{-1} \in C_{diff}^\infty(\hat{U}_j, \hat{U}_i)$$

And for every smooth charts, there exists a **maximal chart** which contains the smooth charts. Such atlas is called the **smooth structure** of M .

◇ **Lemma 1.1.0.2** Let M be a topological manifold:

- Every smooth atlas is contained in a maximal smooth atlas;
- Two smooth atlases for M determines the same maximal smooth atlas iff their union is smooth.

◦ **Proof**

- Let $\mathcal{A}$ be the collection of charts that are smooth compatible with every chart in $\bar{\mathcal{A}}$.
 $\bar{\mathcal{A}}$ is a smooth chart. Firstly, $\mathcal{A} \subseteq \bar{\mathcal{A}}$. For two charts $(U, \varphi), (V, \psi) \in \bar{\mathcal{A}}$ where $U \cap V \neq \emptyset$, select some $p \in U \cap V$ and a chart (W, θ) where $p \in U \cap V \cap W$. Omitted.
- Omitted.

□

► **Theorem** (Milnor, 1956) S^7 has more than one the smooth structures.

► **Theorem** (Exotic $\mathbb{R}^4$) $\mathbb{R}^4$ has infinite many of smooth structures.

1.1.1 Smooth Functions

Definition 1.1.1.1 A function $f : M \rightarrow \mathbb{R}$ is **smooth function** if for every coordinate (U_i, φ_i)

$$f \circ \varphi_i^{-1} : \hat{U}_i \rightarrow \mathbb{R}$$

is smooth.

1.1.2 Construction of Smooth Manifold

◇ **Lemma 1.1.2.1** (Smooth Manifold Construction Lemma) Let M be a set, and suppose we are given a collection $\{U_\alpha\}$ of subse of M , together with injective maps $\varphi_\alpha : U_\alpha \rightarrow \mathbb{R}^n$. With following properties:

- For each α , $\varphi_\alpha(U_\alpha)$ is an open subset of $\mathbb{R}^n$;
- $\varphi_\alpha(U_\alpha \cap U_\beta)$ are open in $\mathbb{R}^n$;
- Whenever $U_\alpha \cap U_\beta \neq \emptyset$, $\varphi_\alpha \circ \varphi_\beta^{-1}$ is a diffeomorphism;
- Countably many U_α cover M ;

- Hausdorffness;

◦ **Proof** Omitted. □

1.2 Smooth Functions and Smooth Maps

1.2.1 Definitions

Definition 1.2.1.1

- A function $f : M \rightarrow \mathbb{R}^k$ is **smooth** if for every $p \in M$, there exists a smooth chart (U, φ) and the composition $f \circ \varphi^{-1} : \hat{U} \rightarrow \mathbb{R}^k$ is smooth;
Such $f \circ \varphi^{-1}$ is called the **coordinate representation** of f ;
- Furthermore, $F : M \rightarrow N$ is smooth if for some charts $(U, \varphi), (V, \psi)$,

$$\varphi \circ F \circ \psi^{-1} : \psi(U \cap V) \rightarrow \varphi(U \cap V)$$

is smooth

It is easy to verify:

- $C^\infty(M)$ is commutative $\mathbb{R}$ -algebra;
- For every charts, the representation map is smooth;

◇ **Lemma 1.2.1.2** (Smoothness is Local) Let M, N be two smooth manifolds, $F : M \rightarrow N$ be a map. If every point $p \in M$, restriction $F|_U$ is smooth, then F is smooth; Conversely also holds.

◦ **Proof** Omitted; □

◇ **Lemma 1.2.1.3** (Pasting) If given $F_\alpha : U_\alpha \rightarrow N$ with $F_\alpha|_{U_\alpha \cap U_\beta} = F_\beta|_{U_\alpha \cap U_\beta}, \cup U_\alpha = M$. Then there exists $F : M \rightarrow N$ such that

$$F|_{U_\alpha} = F_\alpha$$

◦ **Proof** ... □

◇ **Lemma 1.2.1.4** Every smooth map between smooth manifolds is continuous.

◦ **Proof** Omitted. □

1.2.2 Diffeomorphisms

Definition 1.2.2.1 ...

1.3 Smooth Covering Maps

1.4 Proper Maps

Definition 1.4.0.1 $F : M \rightarrow N$ is **proper** if the inverse image of any compact $K \subseteq N$ is compact.

◇ **Lemma 1.4.0.2** Suppose M is a compact space and N is a Hausdorff space, then every continuous map $F : M \rightarrow N$ is proper.

◦ **Proof ...**

□

1.5 Partition Unity

Chapter 2 Tangent Spaces

2.1 Tangent Vectors

2.1.1 Derivative on Opens

Definition 2.1.1.1 ...

2.1.2 Formel Derivations

2.1.3 Tangent Vectors on a Manifold

♣ Example 2.1.3.1 (Tangent Space of Vector Space) Let $V \cong \mathbb{R}^n$ be a vector space. We have

$$T_w V = V; w \in V$$

2.1.4 Pushforwards

2.2 Coordinate Computations

2.3 Tangent Vectors to Curves, Fibres

2.3.1 Curves

2.3.2 Fibres

Chapter 3 Vector Fields

3.1 Tangent Bundles

Definition 3.1.0.1 Define: For manifold M , the **tangent space**

$$TM = \coprod_{p \in M} T_p M$$

Then:

♠ **Proposition 3.1.0.2** (Smooth Structure of TM , Smoothness of Bundle Map)

◦ **Proof ...**

□

3.1.1 Functoriality

♠ **Proposition 3.1.1.1** $T : Man^1 \rightarrow VBun$ is a functor.

◦ **Proof**

- Firstly, T is well-defined;
- Secondly, $T(g \circ f) = T(g) \circ T(f)$. $f : M \rightarrow N, g : N \rightarrow L$,

$$\begin{aligned} T_p(g \circ f)(X)(h) &= X(h \circ (g \circ f)) \\ &= T_p(f)(X)(h \circ g) \\ &= T_p(g) \circ T_p(f)(X) \end{aligned}$$

where $X \in T_p M$.

□

Especially

♠ **Proposition 3.1.1.2** If $f : M \rightarrow N$ is smooth map, so is $T(f)$.

◦ **Proof** Consider followin diagram

$$\begin{array}{ccc} TM & \xrightarrow{T(f)} & TN \\ \downarrow & & \downarrow \\ M & \xrightarrow{f} & N \end{array}$$

Take charts

□

3.1.2 Pfaffian Form

Definition 3.1.2.1 Let M be a differentiable manifold and V a finite dimensional vector space. A **V -valued Pfaffian form**, also called 1-form on M is a smooth map

$$\omega : TM \rightarrow V$$

where $\omega_p : T_p M \rightarrow V$ is linear.

The set of all Pfaffian forms on M

$$\Omega^1(M, V)$$

Especially, when $V = \mathbb{R}$, denoted by $\Omega_1(M)$.

Definition 3.1.2.2 Define the product

$$\begin{aligned} C^\infty(M) \times \Omega^1(M, V) &\rightarrow \Omega^1(M, V) \\ f \cdot \omega &:= (f \circ \pi_{TM}) \cdot \omega \end{aligned}$$

♣ **Example 3.1.2.3** ...

Pullback

Definition 3.1.2.4 Let $f : N \rightarrow M$. Then, the **pullback**

$$\begin{aligned} f^* : \Omega^1(M, V) &\rightarrow \Omega^1(N, V) \\ \omega &\mapsto \omega \circ T f \end{aligned}$$

3.2 Lie Algebra of Vector Fields

◇ **Lemma 3.2.0.1** If $U \subseteq M$ is an open subset and $f \in C^\infty(U, V)$ is a smooth function on U with values in V , where V is a finite dimensional vector space, $X \in \mathcal{V}(M)$.

Then we obtain for each a smooth function

$$\mathcal{L}_X f := df \circ X|_U \in C^\infty(U, V)$$

◦ **Proof** Omitted. □

Definition 3.2.0.2 Such $\mathcal{L}_X(f)$ is called **Lie derivative** of f with respect to X .

3.2.1 Lie Bracket

◇ **Lemma 3.2.1.1** Let $U \subseteq \mathbb{R}^n$ be an open subset. Then we obtain a Lie bracket on the space $C^\infty(U, \mathbb{R}^n)$ by

$$[X, Y](p) := dY(p)X(p) - dX(p)Y(p); p \in U$$

The map

$$C^\infty(U, \mathbb{R}^n) \rightarrow \text{End}(C^\infty(U, \mathbb{R})), X \mapsto \mathcal{L}_X$$

where $\mathcal{L}_X(f)(p) = df(p)X(p)$. is an injective homomorphism of Lie algebras. i.e. $\mathcal{L}_{[X, Y]} = [\mathcal{L}_X, \mathcal{L}_Y]$

◦ **Proof** ... □

3.3 Vector Field as Derivations

Chapter 4 The Cotangent Bundle

4.1 Tangent Covectors on Manifolds

Definition 4.1.0.1 Let M be a manifold. For each $p \in M$, we define the **cotangent space** at p ,

$$T_p^* M := (T_p M)^\vee$$

4.2 Differentials

Definition 4.2.0.1 ...

Chapter 5 Submersions, Immersions and Embeddings

Chapter 6 Submanifolds

6.1 Embedded Submanifold

♠ **Proposition 6.1.0.1** Suppose $F : M \rightarrow N$ smooth, $q \in N$ regular value. Then, for $p \in f^{-1}(q)$,

$$T_p(f^{-1}(q)) = \text{Ker}(d_p F)$$

6.2 Immersed Submanifold

Definition 6.2.0.1 Immersed submanifold: Image of injective immersion.

♣ **Example 6.2.0.2** $\mathbb{R} \rightarrow T, x \mapsto (\exp(2\pi ix), \exp(2\pi irx))$.

Fact If f is closed injective immersion then $\text{Im}(f)$ is an embedded submanifold.

Chapter 7 Differential Forms

7.1 Differential Form

7.1.1 title

Here is some of review of

♠ **Proposition 7.1.1.1** We have

$$\begin{cases} \dim \Lambda(V) = 2^d \\ \dim \Lambda_k(V) = \binom{d}{k}, 0 \leq k \leq d \end{cases}, \dim(V) = d, \Lambda_k = \{0\}; k \geq d$$

◦ **Proof ...**

□

Univeresal Mapping Property

♠ **Proposition 7.1.1.2** Denote $\varphi : V \times \cdots \times V \rightarrow \Lambda_k(V), (v_1, \dots, v_k) \mapsto v_1 \wedge \cdots \wedge v_k$. Then, for arbitrary alternating multilinear map h of $V \times \cdots \times V$ into the vector space W . There exists a unique map $\tilde{h} : \Lambda_k(V) \rightarrow W$ such that

$$\tilde{h} \circ \varphi = h$$

With the following diagram commutes

$$\begin{array}{ccc} & \Lambda_k(V) & \\ \varphi \uparrow & \searrow & \\ V \times \cdots \times V & \xrightarrow{h} & W \end{array}$$

◦ **Proof ...**

□

★ **Corollary 7.1.1.3** Suppose that $W = F$. Then, we could take the notion of

$$\Lambda_k(V)^* \cong A_k(V)$$

◦ **Proof ...**

□

Pairing

Definition 7.1.1.4 Let V and W be real finite dimensional vector spaces, a **pairing** of V and W is a bilinear map

$$(-, -) : V \times W \rightarrow \mathbb{R}$$

A **pairing** is called **non-singular** if whenever $w \neq 0$ in W , there exists

Linear Transformation of $\Lambda(V)$

Take the notion

$$\text{End}(\Lambda(V))$$

for the endomorphisms of $\Lambda(V)$

7.1.2 Tensor Fields and Differential Forms

7.1.3 The Lie Derivative

7.2 Orientation

7.2.1 Orientation of Vector Spaces

Definition 7.2.1.1 Let V be an N -dimensional real inner product space. We extend the inner product from V to all of $\Lambda(V)$

$$\langle w_1 \wedge \cdots \wedge w_p, v_1 \wedge \cdots \wedge v_p \rangle := \det \langle w_i, v_j \rangle$$

The bilinear map can be extended linearly to $\Lambda_p(V)$.

♠ **Proposition 7.2.1.2** If $e_1, \dots, e_n$ is an orthonormal basis of V , then the corresponding basis of $\Lambda(V)$ is an orthonormal basis for $\Lambda(V)$.

◦ **Proof ...**

□

Definition 7.2.1.3 (Consistently Oriented) Assume that $(v_1, \dots, v_n)$ and $(w_1, \dots, w_n)$ be two bases of V with $\dim(V) = n$, then we have the transform matrix

$$v^T = Bw^T$$

Then, they are **consistently oriented** if $\det(B) > 0$.

7.2.2 Orientation of Manifold

Definition 7.2.2.1 Let M be a manifold. Then a frame

$$(E_1, \dots, E_n)$$

is **(positively) oriented** if

7.2.3 The Orientation Covering

7.2.4 The Orientation of Hypersurface

7.2.5 The Orientation of

Chapter 8 Integration on Manifolds

Definition 8.0.0.1 Let M be a manifold. A **partition of unity** is a collection of smooth maps $(\varphi_i : M \rightarrow \mathbb{R})$

- there exists $\varphi_i : U_i \rightarrow V_i$ such that

$$\text{supp}(\varphi_i) \subseteq U_i$$

- For each $p \in M$, only finitely many of $\varphi_i(p) \neq 0$.

Definition 8.0.0.2 Let M be an oriented n -dim smooth manifold, ω a diff n -form, local integrable. Choose a partition of unity (φ_i) , define

$$\int \omega = \sum_i \int h_i^{-1*}(\varphi_i \circ \omega)$$

Chapter 9 De Rham Cohomology

Chapter 10 The de Rham Theorems

Chapter 11 Integral Curves, and Flows

11.1 Integral Curves

Definition 11.1.0.1 ...

Chapter 12 Integral Manifolds and Foliations

Chapter 13 The Hodge Theorem